

The empirical recurrence for OEIS A304469, recovered from its tail

Adrian Perez Fontelles

Independent researcher

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Abstract

OEIS A304469 records “Empirical recurrence of order 84” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 134$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 5$, so the recurrence is proved for every $n \geq 89$. Its last coefficient is $c_{84} = 13312 \neq 0$, so no further tail satisfies anything shorter; any order-84 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A304469 is “Number of $n \times 5$ 0..1 arrays with every element unequal to 0, 1, 2, 3 or 6 king-move adjacent elements, with upper left element zero.”. It has offset 1 and begins

16, 120, 316, 1100, 5063, 18879, 68338, 289227, ...

The entry states:

Empirical recurrence of order 84 (see link above)

As of the “Last modified” line on the live entry (May 13 2018 10:39:43, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 134$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 134$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 84: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 5$ is the first whose minimal order is exactly the stated 84.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 268$ exact terms from $n = 5$ on, it returns order 84 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{84} c_i a(n-i),$$

c_1	10	c_{22}	59493699	c_{43}	−2124246326	c_{64}	−89415907
c_2	−33	c_{23}	176245586	c_{44}	2665987724	c_{65}	−393250864
c_3	93	c_{24}	−173718852	c_{45}	−2629335948	c_{66}	281426756
c_4	−518	c_{25}	−183303337	c_{46}	−90245641	c_{67}	−133847816
c_5	1377	c_{26}	−316842527	c_{47}	251262117	c_{68}	306151694
c_6	−1648	c_{27}	491239862	c_{48}	199926525	c_{69}	−192839094
c_7	8845	c_{28}	402933831	c_{49}	2942258815	c_{70}	95195044
c_8	−25046	c_{29}	275563902	c_{50}	−3044230455	c_{71}	−136617264
c_9	10506	c_{30}	−861884386	c_{51}	2167848089	c_{72}	78105552
c_{10}	−77174	c_{31}	−746208517	c_{52}	−4619748187	c_{73}	−34298400
c_{11}	295120	c_{32}	309301962	c_{53}	3999969447	c_{74}	39885016
c_{12}	−1820	c_{33}	787487492	c_{54}	−2918129012	c_{75}	−20659664
c_{13}	315561	c_{34}	1312175456	c_{55}	4249451616	c_{76}	7433056
c_{14}	−2539725	c_{35}	−1636972798	c_{56}	−2975001810	c_{77}	−7515296
c_{15}	−29210	c_{36}	278944797	c_{57}	2070113585	c_{78}	3481760
c_{16}	416855	c_{37}	−2120391870	c_{58}	−2581758432	c_{79}	−899936
c_{17}	15370467	c_{38}	3245596671	c_{59}	1226995694	c_{80}	817088
c_{18}	−3569097	c_{39}	−2105284598	c_{60}	−760804610	c_{81}	−334208
c_{19}	−11204949	c_{40}	2707135661	c_{61}	958806711	c_{82}	43776
c_{20}	−63369557	c_{41}	−3938534830	c_{62}	1452436	c_{83}	−36352
c_{21}	36589377	c_{42}	3340823719	c_{63}	−69400947	c_{84}	13312

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 89$, and it is the recurrence OEIS A304469 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 5$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{84} - c_1x^{84-1} - \dots - c_{84}$. Its constant term is $-c_{84} = -13312$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 84 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 84, so $\deg q = 84$, and it is monic. It holds from some index t on. From $\max(t, 5)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 84, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 22 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 84, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 13312.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-84 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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